

# Platform-Controlled Trend Propagation in Social Networks: A Reinforcement Learning Approach with Multi-Trend Competition

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## Abstract

We study platform-controlled trend propagation on the Facebook ego network (4039 nodes), modeling multi-trend competition as coupled SIS contagion dynamics with per-node fatigue and a simplex constraint over attention shares. We compare spectral, centrality, frontier, and signed-frontier action decoders for proximal policy optimization (PPO). We find that static spectral and centrality features are weak action-selection signals compared with the dynamic adoption frontier. A boost-only frontier heuristic accelerates spread but cannot regulate low targets due to its inability to suppress existing network dynamics. Introducing signed frontier actions allows PPO to learn target-dependent amplification and suppression, reducing deviation by 60–93% relative to no intervention across target regimes. All paired tests against no-intervention are statistically significant at  $p \leq 2.1 \times 10^{-7}$ . These results demonstrate that effective graph contagion control requires both the right local feature and the right capabilities: the frontier identifies where intervention can propagate, while signed actions enable learned policies to regulate system dynamics.

## 1 Introduction

Social media platforms shape which content reaches users through algorithmic curation. Decisions about what to amplify, demote, or distribute affect the trajectory of trends in ways that influence user engagement, public discourse, and platform revenue. Understanding how to design intervention strategies on social networks is therefore both practically and theoretically important, but the question of which signals are most useful for guiding intervention remains open.

This paper models platform-controlled trend propagation as a reinforcement learning problem on a real social network. We model multi-trend competition for finite user attention as coupled SIS contagion dynamics with per-node fatigue. Each node distributes its attention across several trends under a simplex constraint, and the platform agent boosts or suppresses specific nodes to steer collective attention toward a target share for one designated trend. The dynamics extend standard SIS contagion with fatigue feedback that affects both adoption susceptibility and broadcast capacity.

A central design question is how the agent should choose which nodes to intervene on. Direct per-node action spaces scale poorly with network size and produce policies that memorize node

identities rather than learning transferable strategies. We instead use weights over precomputed node signals: spectral modes of the graph Laplacian, centrality measures, and dynamic frontier scores. The environment computes per-node intervention scores from these weights and applies budget-constrained boosts or suppressions.

Our findings unfold in two stages. First, decoders relying on global topological features underperform those using dynamic frontier signals, the moving boundary between adopted and susceptible nodes. Second, when the action space supports both amplification and suppression supported through the frontier signal, PPO achieves substantial improvement over heuristic baselines. The boost-only experiment identifies the frontier as the correct node-selection signal; the signed-frontier experiment shows that PPO adds value by learning target-dependent direction and intensity over that signal.

The main parts are: (a) a multi-trend SIS contagion model with attention competition and per-node fatigue; (b) a decoder ablation methodology for graph-based RL action spaces; (c) empirical evidence that for SIS-style contagion control, dynamic exposure signals dominate static topology under node-level observability; (d) demonstration that signed-action policies, learned via PPO, achieve statistically significant precision homeostatic control where heuristics cannot.

## 2 Background

### 2.1 Contagion Dynamics on Networks

Standard SIS (susceptible-infected-susceptible) models describe contagion as a balance between infection from adopted neighbors and recovery to a non-adopted state. Pastor-Satorras and Vespignani (2001) analyzed SIS dynamics on scale-free networks and showed that the absence of an epidemic threshold (spreading rate that leads to dying out) in such networks allows for even weak infectious agents to persist. Their work established that network topology, particularly degree heterogeneity, fundamentally shapes propagation dynamics. This result is foundational for our problem because the Facebook ego network used in this work is highly heterogeneous, so the contagion regime is strongly influenced by degree variation and local network structure.

Their analysis is limited for our purposes in three ways. First, it considers a single contagion in isolation rather than competing trends. Second, it assumes constant infection and recovery rates, which misses real engagement patterns where users tire of overexposure. Third, it does not address active platform intervention. Our model extends their framework by adding multi-trend competition, per-node fatigue with heterogeneous archetype parameters, and a reinforcement learning controller.

### 2.2 Generative Models for Centrality Distributions

Chebolu and Melsted (2008) analyzed two random graph models for the web, showing equivalence between a random surfer construction and PageRank-based selection. Their key result is that the expected in-degree and PageRank of vertices both follow power laws with the same exponent under the random surfer dynamics, with a phase transition at parameter value  $p = 1/2$  where the root vertex undergoes different behavior than non-root vertices.

This work informed an exploratory direction in our project: using synthetic graph models with controlled centrality distributions to study how topology affects RL controllability. Eventually allowing for meta-policy learning for multi-trend control. One possible route is a hierarchical sampling framework in which one centrality measure is treated as a parameter governing the conditional distribution of others, with bifurcation analysis on the resulting parameter functions used to characterize regime changes. Because the final experiments focused on learned intervention policies on the empirical Facebook network, this generative direction remains outside the scope of the present work.

### 3 Methodology

#### 3.1 Network Data

We use the Facebook social circles dataset from the Stanford Network Analysis Project (Leskovec and Krevl, 2014). The combined ego network contains 4039 nodes and 88,234 undirected edges, with average degree 43.69, average clustering coefficient 0.606, diameter 8, and a single connected component. The network exhibits modular structure with several distinct communities visible in spectral embeddings, providing realistic topology for studying multi-trend competition.

#### 3.2 Multi-Trend Contagion Dynamics

Each node  $i$  has attention shares  $V_{i,t} \in [0, 1]$  across  $T$  trends, satisfying the simplex constraint  $\sum_t V_{i,t} = 1$ . Trend-specific spreading strength is represented by a global spreading rate  $\beta$  and a trend-specific strength multiplier  $s_t$ , so that  $\beta_t = \beta \cdot s_t$ . This allows us to easily differ trend propagation strength while maintaining a shared baseline infection structure.

The dynamics for each trend follow:

$$\frac{dV_{i,t}}{dt} = \beta_t \cdot a_{i,t} \cdot \sigma_i \cdot (1 - V_{i,t}) \cdot \sum_j A_{ij} E_{j,t} V_{j,t} - \Phi_{i,t} V_{i,t} \quad (1)$$

Each term has a specific role:

- $\beta_t = \beta \cdot s_t$  is the effective propagation rate for trend  $t$ , combining a global spreading rate with a trend-specific strength multiplier.
- $a_{i,t}$  is the affinity of node  $i$  for trend  $t$ , computed from the node interest vector and trend embedding. High affinity means the node is receptive to this specific trend.
- $\sigma_i$  is the conductivity of node  $i$ , a static archetype property determining how strongly the node broadcasts adopted content.
- $(1 - V_{i,t})$  is the susceptibility factor: nodes can only adopt up to saturation, mirroring standard SIS formulations.

- $\sum_j A_{ij} E_{j,t} V_{j,t}$  is the infection pressure from neighbors, weighted by their effector quality and current trend share.
- $\Phi_{i,t} V_{i,t}$  is fatigue-driven decay: more fatigue causes faster decay of the corresponding trend share.

The effector quality  $E_{j,t} = \sigma_j a_{j,t} / (1 + \Phi_{j,t})$  captures the node’s effective broadcast strength. A conductor with high affinity and low fatigue projects strong influence; a fatigued node projects weak influence regardless of baseline conductivity.

Fatigue evolves per trend with recovery and accumulation:

$$\frac{d\Phi_{i,t}}{dt} = -\rho_i \Phi_{i,t} + \eta V_{i,t} \quad (2)$$

where  $\rho_i$  is the per-node recovery rate and  $\eta$  is a global accumulation constant.

After each Euler step, we renormalize each row of  $V$  to maintain the simplex constraint. This enforces the conservation principle that user attention is finite.

### 3.3 Node Properties and Archetypes

Each node receives independent random assignment to one of four parameter combinations formed by two binary axes: conductivity and fatigue recovery. Conductivity is either high, with  $\sigma_i = 1.0$ , or low, with  $\sigma_i = 0.3$ . Fatigue recovery is either fast, with  $\rho_i = 0.15$ , or slow, with  $\rho_i = 0.05$ . For simplicity, we refer to these four parameter combinations as interpretive archetypes, though they arise directly from the underlying conductivity and fatigue parameters rather than being separate model components. The four archetypes are: high conductivity and slow fatigue recovery, high conductivity and fast fatigue recovery, low conductivity and slow fatigue recovery, and low conductivity and fast fatigue recovery.

Each node has an interest vector  $u_i \in R^K$  sampled uniformly on the unit sphere. Each trend has an embedding  $v_t \in R^K$  similarly sampled. Affinity is computed as  $a_{i,t} = (u_i^\top v_t + 1)/2$ , mapping cosine similarity from  $[-1, 1]$  to  $[0, 1]$ . This is done under the assumption that users that have low affinity with a trend will not actively demote it just not interact with it. Both vectors are static throughout an episode. While real users evolve their interests over time and trends drift in semantic content, we hold these fixed to isolate the effects of network dynamics, competition, and fatigue.

### 3.4 Action Space and Decoder Variants

Our action parameterization deliberately avoids the most obvious choice: a binary vector of length  $n$  specifying which nodes to boost. While such per-node actions are expressive, they create two pathologies for learning. First, the policy must learn 4039 independent decisions from an aggregate reward signal, producing high-variance gradients that converge slowly. Second, the policy can memorize node identities rather than learning generalizable selection rules. A 4039-dimensional output trained on a fixed graph effectively encodes “node 1247 is useful” rather than “nodes with

property X are useful.” This memorization fails if the graph topology, trend embedding, or seeding configuration changes.

We instead parameterize actions as a fixed-size vector independent of network size:

$$\text{action} = [w_1, \dots, w_K, w_{\text{hub}}, w_{\text{auth}}, w_{\text{deg}}] \in [-1, 1]^{K+3} \quad (3)$$

The first  $K$  entries weight Laplacian eigenvectors, representing spectral modes of the graph. The remaining three weight centrality features. The environment interprets this action through a decoder that determines per-node intervention scores. Top- $k$  nodes by absolute score receive signed boost or suppression within a budget constraint, depending on the decoder.

We tested six decoder variants:

1. **current**: multiplicative composition of spectral and centrality preferences, modulated by frontier-based boost effectiveness
2. **additive**: linear combination spectral + centrality + frontier
3. **spectral\_frontier**: projects the frontier signal into the Laplacian basis, reconstructs a smoothed frontier pattern, and combines it with centrality
4. **frontier**: centrality-weighted dynamic frontier signal
5. **raw\_frontier**: pure frontier signal  $f_i = (1 - V_{i,\text{target}}) \cdot (AV_{\text{target}})_i$ , boost-only
6. **signed\_frontier**: action vector reduced to scalar direction  $d = \text{mean}(\text{action})$ . If  $d > 0$ , boost top- $k$  frontier. If  $d < 0$ , suppress top- $k$  high-share, high-fatigue target-trend nodes

The frontier signal  $f_i$  measures how much non-adopted attention at node  $i$  is exposed to neighbors currently engaged with the target trend. Essentially identifying the local boundary where contagion can productively spread.

### 3.5 Reward and Training

The reward penalizes deviation from target share, accumulated fatigue, and intervention energy:

$$R_t = -|V_{\text{target}}(t) - V^*| - \lambda_{\Phi} \bar{\Phi}_{\text{target}}(t) - \lambda_E E(t) \quad (4)$$

We train PPO using stable-baselines3 with default hyperparameters for 150,000 timesteps, with episodes of 100 steps. Comparison baselines include no-intervention, random action, greedy degree-based selection, the boost-only raw frontier heuristic, and a target-aware frontier heuristic that boosts when current share is below target and suppresses when above. All evaluation metrics are computed as means and standard deviations over 10 evaluation episodes with distinct random seeds.

## 4 Results

### 4.1 Single-Trend Dynamics Validate the SIS Model

Without intervention, the single-trend system exhibits the canonical epidemic curve: slow growth from the initial seed set, rapid expansion through exposed neighbors, peak adoption, and eventual decline as fatigue accumulates (Figure 1). Mean adoption peaks near  $t = 70$  at approximately 70% before declining. At peak, approximately 84% of nodes satisfy  $V > 0.5$ , confirming that the SIS-with-fatigue formulation produces realistic propagation and saturation dynamics on the Facebook network.

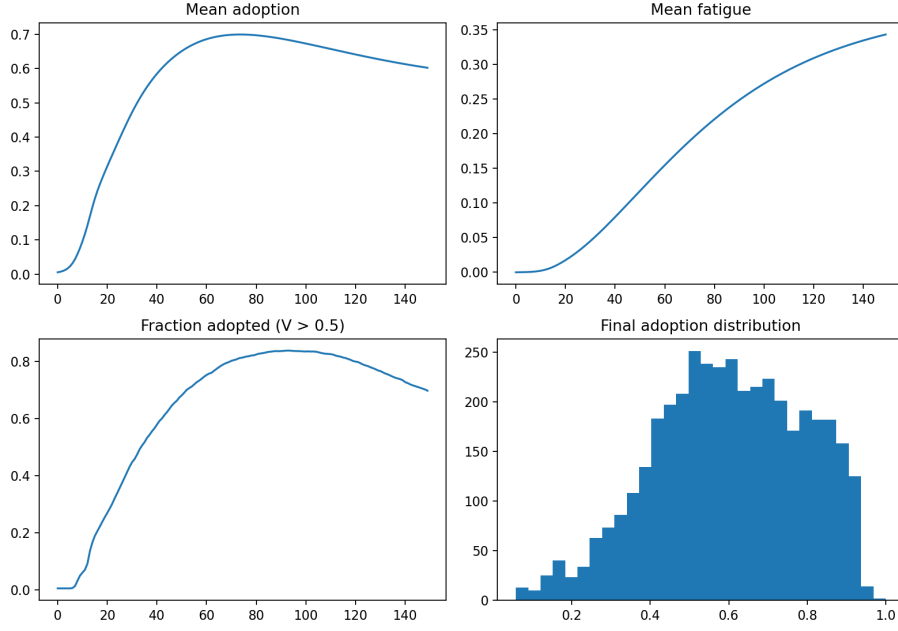


Figure 1: Single-trend SIS contagion dynamics on the Facebook network with fatigue. Mean adoption rises sigmoidally, peaks around  $t = 70$ , and declines as fatigue accumulates.

### 4.2 Energy-Cost Tradeoff in Single-Trend Control

We swept the energy penalty weight  $\lambda_E$  from 0 to 0.10 to characterize the tradeoff between control quality and intervention cost (Figure 2). PPO achieves minimal adoption-deviation improvements over no-intervention, reducing deviation by roughly 3–7% depending on the energy penalty. However, the reward gain over no-intervention is small in the single-trend setting, and PPO falls behind no-op once the energy penalty becomes large enough for intervention costs to dominate.

The Pareto curve shows that  $\lambda_E = 0.05$  gives the best observed deviation while reducing intervention energy by approximately 4.7% relative to unconstrained training. This suggests that single-trend control has limited room for improvement because the natural trajectory already passes near the target regime; aggressive intervention often adds fatigue and energy cost without substan-

tially improving adoption error.

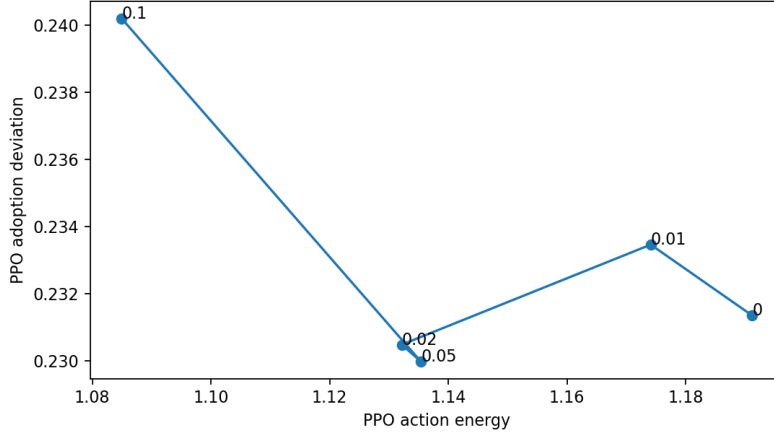


Figure 2: Pareto frontier between adoption deviation and intervention energy across energy penalty weights for single-trend control at target=0.30.

### 4.3 Multi-Trend Asymmetric Competition

With four trends using strength multipliers  $s = [1.0, 1.25, 1.0, 0.75]$  for the background, strongest competitor, equal competitor, and disadvantaged target trend, respectively, the natural dynamics produce a clear ordering (Figure 3). Under no intervention, final mean shares are: trend 0 = 0.264, trend 1 = 0.322, trend 2 = 0.250, and trend 3 = 0.165. Across 10 seeds, trend 3 final share is  $0.165 \pm 0.002$ .

Trend 1 dominates because of its higher strength multiplier, while trend 3 remains disadvantaged because of its lower strength. Setting the target share for trend 3 to 0.30 creates a natural deviation gap of approximately 0.135 that no-intervention cannot close.

### 4.4 Decoder Ablation: Topology vs Frontier Features

The decoders compared here all lack suppression capability. By restricting all decoders to boost-only, we isolate the effectiveness of node-selection signals independently of directional control

Table 1 shows mean reward, adoption deviation, and intervention energy across five decoders for target trend 3 with target share 0.30.

Three patterns emerge. First, decoder selection produces substantial variation in PPO performance, with reward ranging from  $-23.44$  for the additive decoder to  $-20.51$  for the raw\_frontier decoder. Second, decoders incorporating frontier signals consistently outperform those relying on global topology alone. Third, in this boost-only setting, the raw frontier heuristic ( $-19.50$ ) outperforms all learned policies, including the best PPO configuration. This indicates that while learning can approximate useful behavior, the frontier already provides the best node-selection signal.

This result shows concretely that knowing the structural properties of the network is unreliable when it comes to promoting trends. This confirms that static structural properties are insufficient

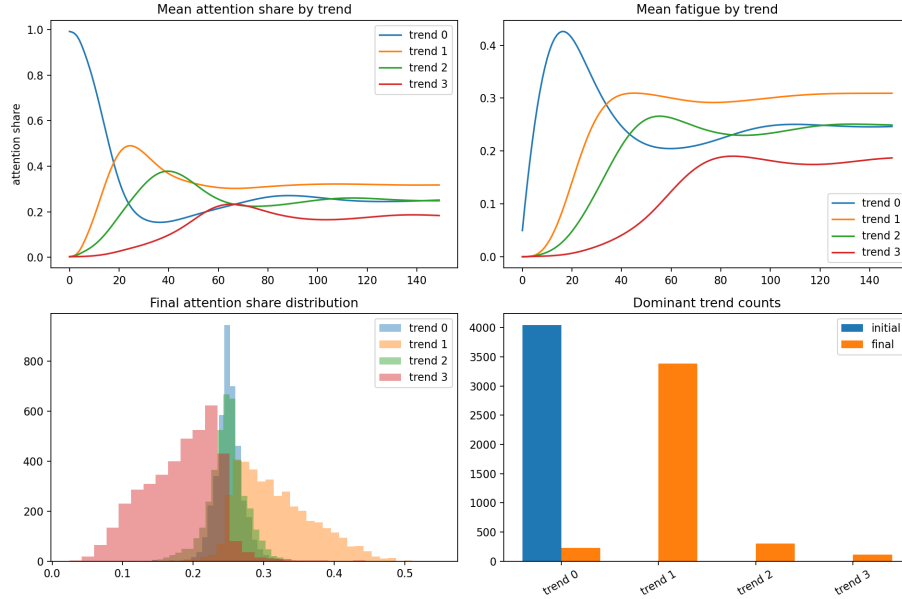


Figure 3: Natural multi-trend dynamics with asymmetric trend strengths. Trend 1 dominates; trend 3 plateaus at approximately 16.5% share, providing a meaningful control gap when the target is 30%.

for selecting intervention targets; the critical signal is to know where the current contagion boundary lies.

#### 4.5 Boost-Only Control Regime Characterization

To understand when frontier intervention helps versus hurts under boost-only mechanisms, we sweep target share from 0.20 to 0.60 with the raw frontier policy held constant (Table 2).

The frontier policy reaches a natural ceiling of approximately 0.33 share regardless of target. Below this ceiling, as in the target=0.20 case, the policy overshoots and reduces reward by 19.77 relative to no-intervention. Near the ceiling, as in the target=0.30 and target=0.40 cases, the policy provides modest to strong improvement. Above the ceiling, as in the target=0.50 and target=0.60 cases, the policy provides maximum improvement but cannot reach the target.

This analysis reveals a property of boost-only policies: their effective ceiling is determined by network topology and dynamics, not by the controller’s intent. Achieving precise control near or below this ceiling requires either suppression to prevent overshoot or target-aware stopping criteria.

#### 4.6 Frontier-Mediated Homeostatic Control with Signed Actions

This experiment resolves the limitation identified in the boost-only regime by introducing symmetric control. While the previous analysis showed that the frontier is the correct node-selection signal, it also highlighted the inability to regulate outcomes near or below their natural ceiling. The signed\_frontier decoder extends the action space to include both amplification and suppression, allowing the agent to perform target-dependent regulation.



Table 1: Decoder ablation results on multi-trend control task. Boost-only action mechanism, target trend = 3, target share = 0.30. Mean over 10 evaluation episodes.

| Decoder           | Policy                 | Reward | Deviation | Fatigue | Energy |
|-------------------|------------------------|--------|-----------|---------|--------|
| current           | PPO                    | -22.61 | 0.152     | 0.110   | 0.946  |
| current           | none                   | -21.62 | 0.169     | 0.095   | 0.000  |
| additive          | PPO                    | -23.44 | 0.163     | 0.105   | 0.936  |
| spectral_frontier | PPO                    | -21.35 | 0.144     | 0.116   | 0.559  |
| frontier          | PPO                    | -21.37 | 0.146     | 0.116   | 0.516  |
| raw_frontier      | PPO                    | -20.51 | 0.128     | 0.131   | 0.558  |
| raw_frontier      | raw frontier heuristic | -19.50 | 0.104     | 0.154   | 0.685  |
| raw_frontier      | greedy degree          | -20.98 | 0.137     | 0.123   | 0.543  |

Table 2: Control regime sweep with raw frontier heuristic at fixed intervention parameters. Boost-only action.

| Target | Gap from natural | Reward advantage | Deviation reduction | Frontier final share |
|--------|------------------|------------------|---------------------|----------------------|
| 0.20   | 0.035            | -19.77           | -0.080              | 0.332                |
| 0.30   | 0.135            | -1.53            | 0.102               | 0.332                |
| 0.40   | 0.235            | +7.72            | 0.195               | 0.332                |
| 0.50   | 0.335            | +10.51           | 0.222               | 0.332                |
| 0.60   | 0.435            | +10.70           | 0.224               | 0.332                |

This decoder reduces the action vector to a scalar direction, with positive values corresponding to boost and negative values corresponding to suppression. Node selection within each operation uses fixed frontier-based scoring. Critically, the spectral and centrality components of the action vocabulary are not used for node selection in this decoder. The action’s effect is determined entirely by the frontier-mediated decoder.

Table 3 reports key metrics across three target levels, comparing signed-frontier PPO against no-intervention and a target-aware frontier heuristic that boosts when current share is below target and suppresses when above.

Table 3: Signed-frontier PPO vs baselines across target levels. Mean over 10 episodes. Deviation reduction computed relative to no-op.

| Target | no-op dev. | target-frontier dev. | PPO dev.      | PPO reward | Reduction vs no-op |
|--------|------------|----------------------|---------------|------------|--------------------|
| 0.20   | 0.0787     | 0.0622               | <b>0.0109</b> | -10.18     | 86.2%              |
| 0.30   | 0.1686     | 0.0756               | <b>0.0121</b> | -14.35     | 92.8%              |
| 0.50   | 0.3686     | 0.1527               | <b>0.1461</b> | -30.11     | 60.4%              |

PPO with signed-frontier action reduces deviation by 86.2%, 92.8%, and 60.4% at targets 0.20, 0.30, and 0.50 respectively. Mean action magnitudes scale appropriately with target difficulty: 0.370 at target=0.20, 0.664 at target=0.30, and 1.000 at target=0.50. The agent therefore learns not only whether to amplify or suppress, but also how strongly to intervene based on the target regime.

Table 4 reports paired t-tests confirming statistical significance.

Table 4: Paired t-tests for signed-frontier PPO versus baselines. Same 10 random seeds across conditions.

| Target | Comparison         | Metric    | $t$     | $p$                   |
|--------|--------------------|-----------|---------|-----------------------|
| 0.20   | vs no-op           | reward    | 13.95   | $2.1 \times 10^{-7}$  |
| 0.20   | vs no-op           | deviation | -38.47  | $2.7 \times 10^{-11}$ |
| 0.20   | vs target-frontier | reward    | 35.64   | $5.3 \times 10^{-11}$ |
| 0.20   | vs target-frontier | deviation | -69.82  | $1.3 \times 10^{-13}$ |
| 0.30   | vs no-op           | reward    | 85.37   | $2.1 \times 10^{-14}$ |
| 0.30   | vs no-op           | deviation | -134.24 | $3.6 \times 10^{-16}$ |
| 0.30   | vs target-frontier | reward    | 40.72   | $1.6 \times 10^{-11}$ |
| 0.30   | vs target-frontier | deviation | -62.79  | $3.3 \times 10^{-13}$ |
| 0.50   | vs no-op           | reward    | 137.43  | $2.9 \times 10^{-16}$ |
| 0.50   | vs no-op           | deviation | -227.62 | $3.1 \times 10^{-18}$ |
| 0.50   | vs target-frontier | reward    | 2.04    | 0.072                 |
| 0.50   | vs target-frontier | deviation | -3.07   | 0.013                 |

PPO with signed-frontier action is significantly better than no-intervention on both reward and deviation at every target tested, with all paired tests against no-intervention significant at  $p \leq 2.1 \times 10^{-7}$ . PPO also significantly outperforms the target-aware frontier heuristic at targets 0.20 and 0.30, demonstrating that learning provides value beyond a deterministic target-aware policy. At target 0.50, PPO achieves significantly lower deviation than the target-frontier heuristic ( $p = 0.013$ ), but the reward difference is marginal ( $p = 0.072$ ). This indicates that at high amplification targets where the system operates near its dynamical capacity, learned and heuristic policies converge in overall reward, though the learned policy maintains tighter target tracking.

#### 4.7 Visual Comparison: Heuristic vs. Learned

Figure 4 shows mean target trend share over time under no-intervention versus the raw frontier policy in the default target=0.30 setting. Across 10 seeds, the frontier policy reaches share 0.20 after 23.4 steps on average, while no-intervention requires 53.7 steps, a 2.3x acceleration. Under this boost-only setting, the frontier policy raises final trend 3 share from 0.165 to 0.203.

For visual clarity, Figure 5 shows the stronger target=0.30 frontier intervention regime. The heatmap visualizes per-node trend 3 attention share across the network at five timesteps. By  $t = 25$ , both heuristic and learned frontier policy have established trend 3 presence across multiple communities, while the no-intervention trajectory remains concentrated near the initial seeds.

#### 4.8 Why Frontier Beats Spectral Features

The frontier is entirely local, dynamic, and trend-specific. It identifies nodes that still have available attention for the target trend and are adjacent to neighbors currently carrying that trend. In contagion terms, it locates the active boundary between adopted and susceptible regions of the graph. That is precisely the region where intervention can most productively propagate.

Spectral features have a different character. Low-frequency Laplacian eigenvectors describe

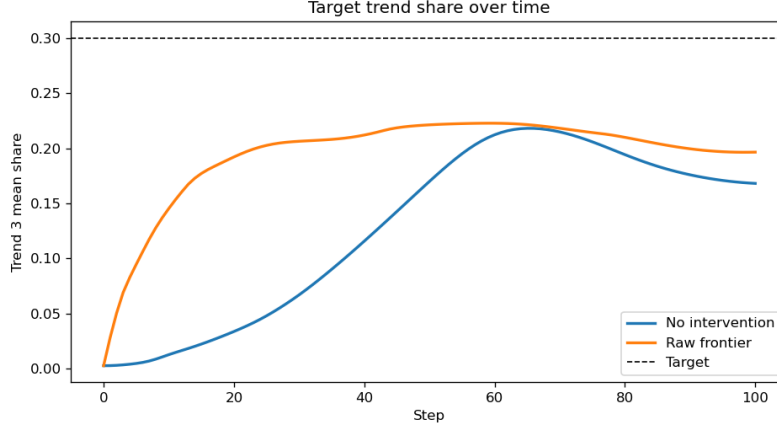


Figure 4: Target trend share over time under no-intervention and raw frontier control. The frontier policy reaches 0.20 target share 2.3x faster on average across 10 seeds.

global, static, and smooth graph structure. They are useful for identifying large-scale geometry, community organization, and smooth variation over the graph. However, the intervention problem studied here depends on sharp, time-varying adoption boundaries. When the adoption state is projected into a spectral basis, the smoothing operation can blur precisely the local boundary information that drives effective intervention.

Centrality features face a similar limitation: they are static. They identify globally important nodes, but do not know whether those nodes are currently exposed, fatigued, saturated, or useful for the target trend. A high-degree node is not automatically a good intervention target if it has already saturated, has little affinity for the target, is highly fatigued, or is not adjacent to active target adoption. Multiplying centrality by spectral scores can therefore mask high-value local frontier nodes whose importance comes from current dynamic position rather than structural prominence.

We do not claim spectral methods are uninformative for control problems generally. Rather, the result establishes that for the specific intervention paradigm studied here, per-node selection with full observability and fixed network topology, the local frontier signal subsumes the structural information that spectral modes provide. Spectral features may remain valuable in regimes where the controller cannot observe every node’s neighborhood state, where intervention budget is severely limited, where actions operate at community rather than node granularity, or where network topology evolves over time.

## 5 Future Work

Several directions warrant investigation. First, per-node policy networks that explicitly observe frontier scores and node features could combine spectral compression in observation with frontier-aware action. Such architectures require permutation-invariant designs, such as shared MLPs over nodes or graph neural networks, but would let the agent modulate intervention intensity based on local context beyond the scalar direction control of `signed_frontier`.

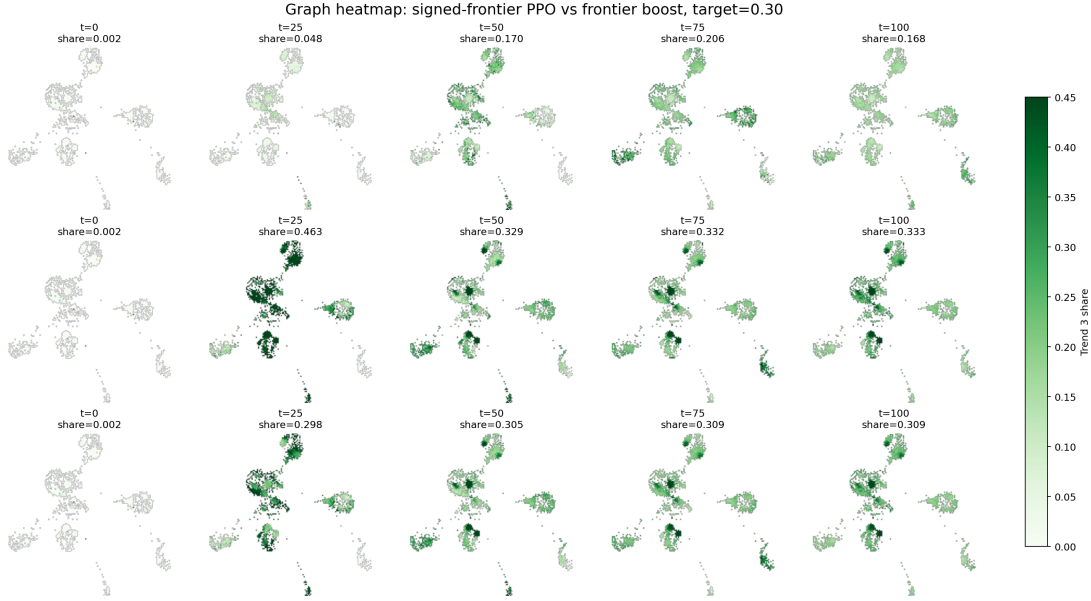


Figure 5: Per-node trend 3 attention share over time in the stronger target=0.30 intervention regime. Top: no-intervention. Middle: raw frontier policy. Bottom: Signed PPO. The frontier policy spreads trend 3 across multiple communities by  $t = 25$ .

Second, spectral methods may become competitive at scales where per-node observation and computation are infeasible. For graphs with millions of nodes, computing frontier scores at every step may become costly, while spectral coefficients of  $V$  can be tracked approximately or incrementally. The smoothing that hurts spectral performance in this setting may become an asset when precise localization is unavailable.

Third, recursive Bellman-style exposure, where influence quality propagates through multiple network hops, would test whether deeper structural features become actionable when the agent can reason over multi-step dynamics. Fourth, multi-target control, where the agent maintains multiple trend shares simultaneously, presents a richer optimization problem than single-target control. Fifth, directed graphs, such as Twitter follower networks, introduce asymmetric influence flows that our undirected formulation cannot capture. Sixth, the hierarchical centrality sampling methodology discussed in the background section offers a path to systematic study of how topology affects controllability through synthetic graph generation with prescribed centrality distributions.

Final addition would be characterizing target feasibility as a dynamical property rather than treating all target shares as equally attainable. In the signed-frontier experiments, high target regimes such as  $V^* = 0.50$  could be approached momentarily but were difficult to sustain, with the target trend relaxing back toward a lower share after fatigue accumulated. This suggests that controllability depends not only on distance from the natural equilibrium, but also on whether the nodes required to sustain the target have sufficient recovery capacity. Future work could therefore estimate a feasibility curve relating target share, intervention budget, fatigue accumulation, and long-run deviation. A natural extension would be recovery-aware frontier control, where the frontier

score is modulated by node recovery rate or current fatigue, for example weighting exposed nodes by  $\rho_i$  or  $(1 - \Phi_{i,t})\rho_i$ . This would test whether higher target shares fail because the controller selects poor sustaining nodes, or because fatigue imposes a structural ceiling on the system.

## 6 Conclusions

We presented a multi-trend SIS contagion model with platform control on the Facebook ego network. Two related findings emerge. First, decoders combining the action vector with global topological features, including Laplacian eigenvectors and centrality measures, underperform decoders using the dynamic frontier signal: the moving boundary between adopted and susceptible nodes. For SIS-style contagion with full local observability, the frontier feature is the dominant local control signal.

Second, learning becomes valuable once the action capability is expressive enough. The boost-only ablation initially suggests that pure heuristics outperform PPO, because a raw frontier heuristic identifies where amplification should occur. However, the signed-frontier extension reveals the deeper structure. Heuristics excel at amplification; learned policies excel at regulation. Pure boost policies hit a one-directional ceiling determined by network dynamics, while symmetric mechanisms enable precision homeostatic control.

PPO with a signed-frontier action space, supporting both amplification and suppression mediated through frontier-based selection, outperforms heuristic baselines across target levels. It achieves 60–93% reduction in deviation from target share relative to no-intervention, with all paired tests against no-intervention significant at  $p \leq 2.1 \times 10^{-7}$ . The learned policy modulates intervention intensity according to target difficulty, applying gentle correction for low targets, moderate effort for reachable targets, and saturated effort for high targets near the system’s dynamical capacity.

Effective graph contagion control therefore required both the right local feature and the right action vocabulary: the frontier identified where intervention could propagate, while signed actions allowed PPO to decide whether to amplify or suppress the trend. The methodological implication is that for graph-based reinforcement learning on contagion-like processes, action feature design matters as much as observation design. Providing the agent with the correct local signal is necessary, but not sufficient; the action space must also express the kinds of control the task requires.

## Acknowledgments

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